

Simulation of vibrational spectra from Argon impurities in deep minerals

M2 SMNO

Alessandro Ciavatta,
supervised by Lorenzo Paulatto
co-supervised by Chrystèle Sanloup

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2.1 Raman spectrum of Corundum
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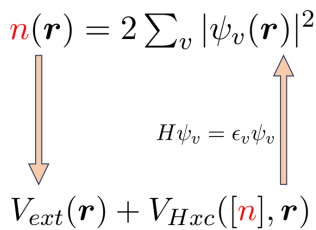
Density Functional Theory (DFT)

Adiabatic approximation: $H_{tot} = H_{ion}(\boldsymbol{R}) + H_{el}(\boldsymbol{r}, \boldsymbol{R})$. Substitutes the electronic solution $E(\boldsymbol{R}) = \langle \Psi_{\boldsymbol{R}} | H_{el} | \Psi_{\boldsymbol{R}} \rangle$, calculated at fixed ionic position $\boldsymbol{R}$

Hohenberg-Kohn: The electronic energy $E = \langle \Psi | H_{el} | \Psi \rangle$ is a functional of the electron density $E[n]$, the ground state is obtained minimizing with respect to density.

Kohn-Sham: Independent electrons with density $n(\boldsymbol{r}) = 2 \sum_v |\psi_v(\boldsymbol{r})|^2$ in a potential $V_{ext}(\boldsymbol{r}) + V_{Hxc}([n], \boldsymbol{r})$

$e^- e^-$ interaction is Hartree + approximated exchange correlation effects.



Forces and 2n+1 theorem

The knowledge of the n-order wavefunction allows the calculation of derivatives of energy up to $(2n+1)$ order, through **perturbation theory** formulas

Forces are properties of unperturbed ground state density in the current configuration: $\frac{\partial E}{\partial \boldsymbol{R}} = -F([n], \boldsymbol{R})$.

For Raman spectroscopy we're interested in: $\frac{\partial^2 E}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J}$ for **phonons**, $\frac{\partial^3 E}{\partial \boldsymbol{\mathcal{E}}^2 \partial \boldsymbol{R}_I}$ for **Raman cross-sections**, $\frac{\partial^3 E}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J \partial \boldsymbol{R}_K}$ for **phonon linewidths**.

$$|\psi_v^{(1)}\rangle = \sum_c |\psi_c^{(0)}\rangle \frac{\langle \psi_c^{(0)} | H^{(1)} | \psi_v^{(0)} \rangle}{\epsilon_v^{(0)} - \epsilon_c^{(0)}}$$

Perturbation Theory

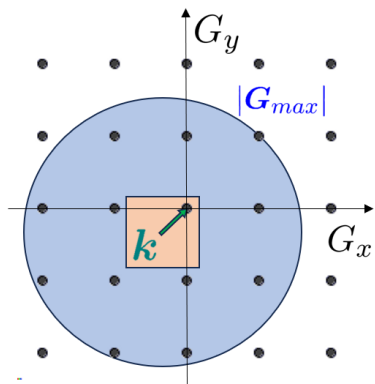
Density Functional Perturbation Theory

$$(H^{(0)} - \epsilon_v^{(0)}) |\psi_v^{(1)}\rangle = -H^{(1)} |\psi_v^{(0)}\rangle$$

Extended systems

Bloch functions: $\psi_{\boldsymbol{k}\nu}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{\boldsymbol{k}\nu}(\boldsymbol{r})$. The periodic part u is expanded in reciprocal vectors plane waves (up to a cutoff $|\boldsymbol{G}_{max}|$ for computational reasons) and the Kohn-Sham Hamiltonians $H_{\boldsymbol{k}} = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} H e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ are diagonalized in a $\boldsymbol{k}$-points grid

The **responses** to perturbations on the Bloch form with different momentum $\boldsymbol{q}$ are **decoupled**: one can construct periodic quantities with no additional complexity for incommensurate $\boldsymbol{q}$



Phonons

Collective vibrations of ions with different wavevector $\boldsymbol{q}$ are **decoupled**, ν indexes the solutions of $m_s\omega^2_{\boldsymbol{q}\nu}\boldsymbol{e}_s(\boldsymbol{q}\nu) = \sum_{s'} C_{ss'}(\boldsymbol{q})\boldsymbol{e}_{s'}(\boldsymbol{q}\nu)$. The basis which describes the displacement switch from the ionic cartesian basis $\boldsymbol{e}_s$ to $\boldsymbol{e}(\boldsymbol{q}\nu)$

The **Interatomic Force Constant** matrix $C_{ss'}(\boldsymbol{q}) = \frac{\partial^2 E}{\partial \boldsymbol{u}_s(-\boldsymbol{q})\partial \boldsymbol{u}_{s'}(\boldsymbol{q})}$ is calculated through a perturbation theory formula from the first-order density $\partial n/\partial \boldsymbol{u}_s(\boldsymbol{q})$

Homogeneous electric fields

Perturbation $-e\boldsymbol{\mathcal{E}} \cdot \boldsymbol{r}$ is ill-defined in periodic systems. Solutions: $\boldsymbol{r} \to d/d\boldsymbol{k}$ or $\langle \psi_m | \boldsymbol{r} | \psi_n \rangle = \frac{\langle \psi_m | [H_{SCF}, \boldsymbol{r}] | \psi_n \rangle}{\epsilon_m - \epsilon_n}$

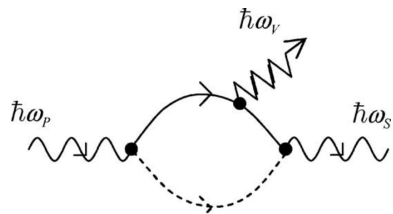
Induced polarization $\boldsymbol{\mathcal{P}}$ depends only **on off-diagonal** elements of $\boldsymbol{r}$, which are well defined. $\chi = \frac{\partial^2 E}{\partial \boldsymbol{\mathcal{E}}^2}$ is calculated from $\partial n / \partial \boldsymbol{\mathcal{E}}$

Phonons induce polarization, response is Born effective charges $Z_s^* = \frac{\partial^2 E}{\partial \boldsymbol{u}_s(\boldsymbol{q}) \partial \boldsymbol{\mathcal{E}}}$. The macroscopic electric field induced by LO phonons in the long wavelength limit splits **LO-TO** degeneracy.

Raman cross-sections

Some modes are **active** for symmetry reasons, phonons are excited at Γ due to conservation of momentum. At lowest order, it's a 1phonon-2photons process.

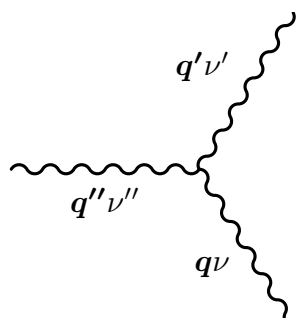
Raman tensor $\frac{\partial^3 E}{\partial \boldsymbol{\mathcal{E}}^2 \partial \boldsymbol{u}_s(\mathbf{0})}$. Mixed derivative, calculated from $\partial n / \partial \boldsymbol{u}_s(\mathbf{0})$ and $\partial n / \partial \boldsymbol{\mathcal{E}}$ from $2n+1$. Alternatively, it's given by $\partial^2 n / \partial \boldsymbol{\mathcal{E}}^2$



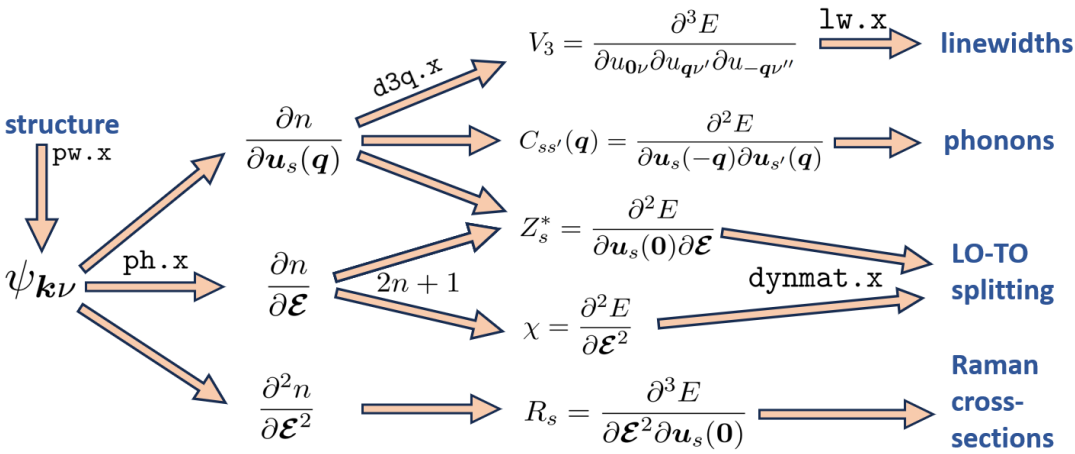
Anharmonicity

Linewidth from **scattering rates**. Transition matrix from third-order anharmonic term, rewritten in the phonon's basis. Linewidth of $\mathbf{0}\nu$ phonon is the result of an integration of $(\boldsymbol{q}\nu', -\boldsymbol{q}\nu'')$ pairs, with energy-momentum conservation

$2n+1$**:** third order derivative w.r.t three phonons needs $\partial n/\partial \boldsymbol{u}$. It's already available after phonon calculation, but it requires a finer mesh of $\psi_{\boldsymbol{k}}$.



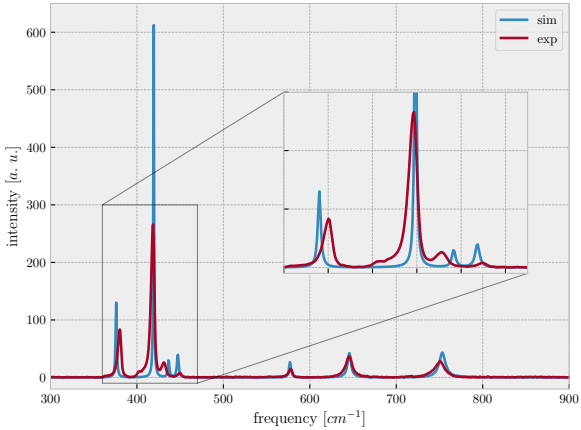
Workflow


structure
pw.x
$\psi_{\boldsymbol{k}\nu}$
ph.x
$\frac{\partial n}{\partial \boldsymbol{u}_s(\boldsymbol{q})}$
$\frac{\partial n}{\partial \boldsymbol{\mathcal{E}}}$
$\frac{\partial^2 n}{\partial \boldsymbol{\mathcal{E}}^2}$
d3q.x
$2n+1$
$V_3 = \frac{\partial^3 E}{\partial u_{\mathbf{0}\nu} \partial u_{\boldsymbol{q}\nu'} \partial u_{-\boldsymbol{q}\nu''}}$
lw.x
linewidths
$C_{ss'}(\boldsymbol{q}) = \frac{\partial^2 E}{\partial \boldsymbol{u}_s(-\boldsymbol{q}) \partial \boldsymbol{u}_{s'}(\boldsymbol{q})}$
phonons
$Z_s^* = \frac{\partial^2 E}{\partial \boldsymbol{u}_s(\mathbf{0}) \partial \boldsymbol{\mathcal{E}}}$
$\chi = \frac{\partial^2 E}{\partial \boldsymbol{\mathcal{E}}^2}$
dynmat.x
LO-TO splitting
$R_s = \frac{\partial^3 E}{\partial \boldsymbol{\mathcal{E}}^2 \partial \boldsymbol{u}_s(\mathbf{0})}$
Raman cross-sections

Raman spectrum of Corundum

Sim. frequencies rescaled by $+0.8\%$
Experimental spectrum: syntethized Corundum, $\lambda_{light} = 780\,\text{nm}$

E. Sorokina et al., *Morphological and chemical evolution of corundum (ruby and sapphire): Crystal ontogeny reconstructed by EMPA, LA-ICP-MS, and Cr3+ Raman mapping*



Ar impurities in Corundum

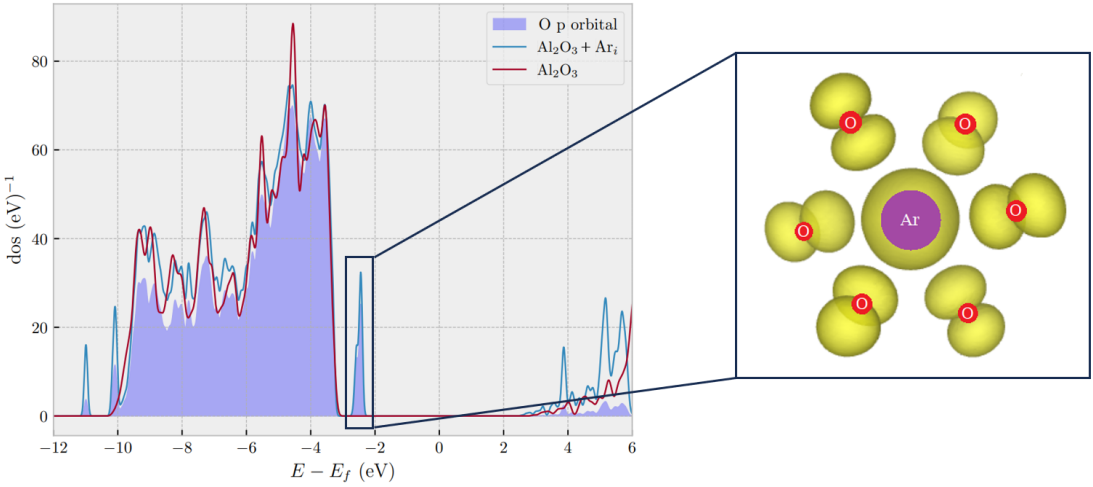
Synthetized at 10.5 GPa, 400 K, measures at standard pressure/temperature. SEM indicates the presence of $\sim 1\,\%$ Ar spreaded homogeneously. XRD data confirms the **presence of interstitials**.

Three possible defects: two substitutions Ar_{Al}, Ar_O and one **interstitial** Ar_i. Defect placed in a $2 \times 2 \times 2$ supercell ($\sim 1\,\%$ Ar). Charged cell and spin-polarized calculations predict metallic states for the substitutions and insulator for interstitial.

Formation Energies

	ΔE (eV)	$\Delta V/V$
Ar_i	11.4	2.9%
Ar_{Al}	16.0	1.8%
Ar_O	17.9	3.8%

Electronic structure of interstitial Ar_i



Raman spectra of interstitial Ar_i

Negative phonon's frequencies for all the structures. Unit and $2 \times 2 \times 2$ cells are highly **perturbed**, while the exp. Raman spectrum does not reveal the presence of Ar

Attempts to fix Ar-p electrons: pseudopotential with **zero valence**, **spin-polarization**, **DFT+U**.

